SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 **COURSE CODE:** Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I KRITHIYA / 192421150 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Krithiya

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	2	2	2	2
Explanation	25%	2	1	1	1
Application	20%	1	1	1	1
Organization	15%	1	1	1	1
Accuracy	10%	1	1	1	1
Clarity	5%	1	1	1	1

Total Marks 28 /40

Signature of the Faculty

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ENT TOOL 2

E: Computer Networks

Limited application weak

No application or incorrect analysis

Poor organization coherent

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Questions

(20)

Before transmission the presentation layer compresses a 900 MB file by 40%. the compressed data is encryption adding 5% overhead to compressed size calculate the compressed file size, the encryption file size and total percentage.

(2) An FTP application transfer 3 GB of data over a network with an effective throughput of 120 Mbps. The Application the data into requests of 3 MB each. Layer divides Calculate the total number of requests generate and estimate transmission time in secs.

(3) A packet experiences the following process delays at diff OSI layers: Application layer = 5 ms, transport = 3 ms, Networks = 4 ms, physical layer = 1 ms. Data link layer = 2 ms propagation delay 18 ms transmission delay = 12 ms Calculate the total end-to-end delay.

(4) A networks transfers 80 MB of useful data using frames of 1600 bytes, where each frames contains 80 bytes of protocol by different OSI layers. Explain how protocol overhead affects network.

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Given:

original file = 900 MB

compression = 40 %

Encryption over ~~loaded~~ headed = 5 % of compressed size.

compressed size :

$$900 \times \frac{60}{100} = 540 \text{ MB}$$

Encryption overhead:

$$540 \times \frac{5}{100} = 27 \text{ MB}$$

Encryption file size :

$$540 + 27 = 567 \text{ MB}$$

Encryption total reduction :

$$\frac{900 - 567}{900} \times 100 = 37\ \%$$

compressed size = 540 MB, Encrypted size = 567 MB, total reduction = 37 %.

The presentation layer compresses data to reduce its size and encrypts it to confidentiality.

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COURSE CODE: Computer Networks

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Annexure A

CONCEPT MAPPING

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Kavitha

htage (%)	Q1 10marks
20%	1
25%	2
20%	2
20%	1
15%	1

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pleted concept map

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2. Given:

Data = 3GB, request size = 3MB, throughput = 120 Mbps.

Total requests

ceiling 1GB = 1000 MB:

$\frac{3000}{3} = 1000$ requests

Transmission time:

3 GB = 3000 × 8 = 24,000 Mb

$\frac{24,000}{120} = 200$ Secs

1000 requests of 200 secs

The Application layer provides services such as file transfer email. & Web access. FTP is used to transfer file between computers.

Given:

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Throughput

Given:

Delay types:

End - to - End Delay:

Application Layer — 4 ms

Transport Layer — 3 ms

Network Layer — 5 ms

Data link Layer — 2 ms

physical Layer — 1 ms

propagation delay — 18 ms

Transmission delay — 12 ms

Adding all the delays.

4 + 3 + 5 + 2 + 1 + 18 + 12

45 ms //

total end - to - end delay = 45 ms

Each layer performs a specific task:

the Application Layer provides user services transport handles end - to - end delivery ! Network handles routing, Data link handles framing and physical

transmit bits processing, transmission and propagation delays together determine the total delay.

4. Frames, overhead and efficiency.

Given:

useful data = 80MB, frame size = 1600 bytes

overhead = 80 bytes per frame.

Assume 1600 bytes is total frame size including overhead.

1600 - 80 = 1520 bytes.

$\frac{80,000,000}{1520} = 52632.58$

52632 × 80 = 4,210,560 bytes

$\frac{1520}{1600} \times 100 = 95\%$

52632 frames

4.21 MB - Approximately overhead

95% - efficiency.

Excessive overhead reduces the proportion of useful data transmitted.